EXPERIMENT 2

Aim: To perform Version Control using GIT

Theory:

Git is a free and open-source distributed version control system (DVCS) used to track changes in files and coordinate work among multiple people. It is fast, lightweight, and supports both small and large projects.

Basic Git Operations:

Initialize – Create a new local repository Add – Stage files for commit Commit – Save changes to local repository Push – Upload local changes to remote repository (GitHub) Pull – Download changes from remote to local

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Practical Steps:

Step 1: Configure Git User

Set up your username and email using the following commands:

git config --global user.name "kunalmotwani223" git config --global user.email "kunalmotwani223@gmail.com"

To verify the configuration:

git config --global --list

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL
$ git config --global --list
user.name=kunalmotwani223
user.email=kunalmotwani223@gmail.com
```

Step 2: Initialize a Repository

Create a project folder and initialize Git inside it:

mkdir git-demo-project cd git-demo-project git init

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/git-demo-project
$ git init
Initialized empty Git repository in C:/Users/Lenovo/Desktop/KUNAL/git-dvcs/git-d
emo-project/.git/
```

Step 3: Commit Changes

Add files to the staging area and commit them:

git add . git commit -m "First Commit"

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/
mo-project (master)
$ git commit -m "First Commit"
[master (root-commit) 34ec911] First Commit
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 index.html
 create mode 100644 teststatus
```

Step 4: View Commit History

Use the log command to view all past commits:

git log --oneline

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/git-de
mo-project (master)
$ git log --oneline
3b6afd6 (HEAD -> master) Express Commit
34ec911 First Commit
```

Step 5: Push to GitHub

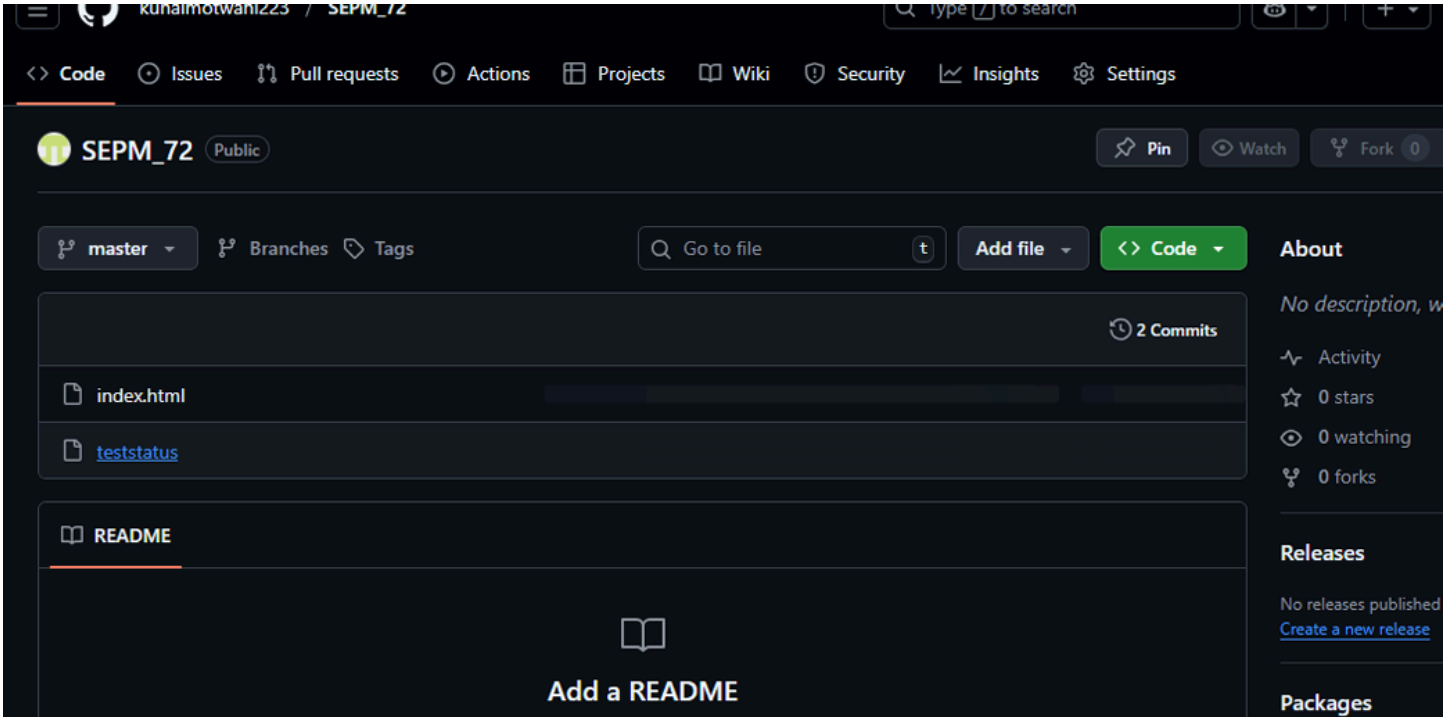
Link your local repository to a remote GitHub repository and push:

git remote add origin https://github.com/kunalmotwani223/SEPM_72.git git push -u origin master

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/git-demo-project (master)
$ git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 28 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 503 bytes | 503.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kunalmotwani223/SEPM_72.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Step 6: Verify on GitHub

After pushing, the files appear on the GitHub repository page.



Step 7: Push Further Changes

Any new changes can be pushed again using:

git push

```
Lenovo@DESKTOP-U1OKVMJ MINGW64 ~/Desktop/KUNAL/git-dvcs/git-demo-project (master)
$ git push
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 28 threads
Compressing objects: 100% (1/1), done.
Writing objects: 100% (2/2), 249 bytes | 249.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kunalmotwani223/SEPM_72.git
   3b6afd6..78a3190  master -> master
```

Conclusion:

Version control using Git was successfully performed. We configured Git, initialized a local repository, committed files, and pushed them to a remote GitHub repository. The entire workflow of tracking and managing code versions was demonstrated successfully.